

OpenSentinel — Sovereign Tech Fund Proposal

Lead Developer: Daniel Raymond Brown

Funding Requested: €100,000

Executive Summary

OpenSentinel is an open-source cybersecurity, encryption, and digital infrastructure initiative designed to protect the open-source ecosystem and public digital networks. It aims to strengthen the resilience of open technologies by developing a suite of freely available tools that detect, prevent, and mitigate security vulnerabilities in critical digital foundations. By offering transparent, community-driven, and sustainable infrastructure, OpenSentinel empowers developers and organizations to secure the global digital commons.

1. Prevalence

OpenSentinel builds upon existing open-source technologies that are deeply embedded in the global digital landscape. Its modular architecture is designed for integration into widely adopted systems such as Linux-based servers, open development frameworks, and secure communication protocols. By supporting tools used by thousands of open-source developers, OpenSentinel directly reinforces the integrity of critical infrastructure relied upon by countless public and private organizations.

2. Relevance

Cybersecurity, encryption, and reliable infrastructure are vital across all major societal sectors — from healthcare to education, journalism, and energy. OpenSentinel contributes to these fields by providing free, accessible, and community-maintained technology that enhances safety and trust in digital systems. It aligns perfectly with the Sovereign Tech Fund's mission to strengthen the open digital base upon which public-interest software depends.

3. Vulnerability

Despite its importance, much of the world's open-source security infrastructure remains underfunded. Volunteers often maintain crucial libraries and tools without adequate resources for long-term stability. OpenSentinel addresses this structural vulnerability by channeling dedicated funding toward sustainable maintenance, security audits, and the development of new defensive modules.

4. Public Interest

OpenSentinel enhances public trust in digital technologies by ensuring that key open-source systems remain secure, transparent, and accessible to all. Its results — open code, documentation, and reusable security frameworks — will directly benefit developers, public institutions, and global communities that rely on open software to deliver essential services.

5. Planned Activities

The project's core activities include: (1) designing and developing an open-source security toolkit; (2) performing independent code audits on critical open-source libraries; (3) establishing best practices for secure infrastructure management; (4) creating educational materials to train open-source developers in threat prevention and mitigation. The 12-month implementation timeline includes milestones for design, testing, community integration, and long-term sustainability planning.

6. Expertise

Daniel Raymond Brown, Lead Developer, brings extensive experience in open-source development, network security, and digital infrastructure. The OpenSentinel project also draws on the expertise of international open-source contributors who share a commitment to maintaining secure, transparent, and freely available software. This collaborative community ensures continuous improvement and resilience.

7. Estimated Costs & Use of Funds

Requested Funding: €100,000 Budget allocation: - 50% — Development and testing of OpenSentinel modules - 25% — Security audits and documentation - 15% — Community outreach, education, and infrastructure - 10% — Administration and compliance The proposed work meets the minimum funding threshold and ensures measurable outcomes that directly benefit the open-source ecosystem.

8. Licensing & Compliance

All OpenSentinel code and documentation will be released under OSI-approved and FSF-recognized licenses (Apache 2.0 for code, CC-BY 4.0 for documentation). These licenses guarantee that all outputs remain free, modifiable, and redistributable, ensuring long-term accessibility and compliance with Sovereign Tech Fund requirements.

Conclusion

OpenSentinel represents a forward-looking commitment to the stability and safety of global open-source infrastructure. It combines technical rigor with a vision of shared digital sovereignty, ensuring that the foundational layers of our connected world remain open, secure, and resilient for generations to come.